

Revisiting and Extending Q-MAML for High-Energy Physics: Classification, Verified VQE Baselines, and Genuine Lattice Hamiltonians

[TODO: author list]
[TODO: affiliation(s)]

Draft (August 23, 2026)

Abstract

We re-test Q-MAML [1] (AAAI 2025), a meta-learned PQC initializer, on HEP problems it was never evaluated on. First, we audit the ML4Sci GSoC 2025 codebase that applies it to HEP classification and find both attempts broken: in one, the training loss is detached from the computation graph before backpropagation, so the learner network never actually updates; the other runs a first-order-MAML-style inner loop, not the paper’s published algorithm. We build a paper-faithful re-implementation and calibrate it against the paper’s own VQE settings at its published 2000-iteration budget (table below). Applied to HIGGS classification, a novel setting for this algorithm, it gets the best validation accuracy at every qubit count tested, significant vs. Gaussian init and the GSoC baseline ($p < 0.05$, $n = 4$ seeds). A McClean-et-al. barren-plateau scan shows no scheme, including ours, changes the variance-scaling exponent; classification cost functions plateau ~ 8 orders of magnitude faster than the VQE costs the paper was validated on. Finally we run ten genuine lattice/field-theory (and adjacent) Hamiltonian pilots (Schwinger model, $\mathbb{Z}_2$ lattice gauge theory, ϕ^4 scalar field, $SU(2)$ lattice gauge theory, collective neutrino oscillations, supersymmetric quantum mechanics, non-Abelian $\mathbb{D}_8$ lattice gauge theory, light nuclei in lattice pionless EFT, scalar Yukawa coupling, and the Lipkin-Meshkov-Glick model): clear wins on six, ties for best on two (every scheme reaches the same final answer, Q-MAML only faster or tied-fastest), a genuine null result on one (too small a system to differentiate any scheme), and one diagnosed loss (gauge invariance enforced by a Gauss-law penalty), plus several corrections to claims made earlier in the paper. A second classification dataset (quark-gluon jet tagging, 2 seeds) replicates the HIGGS win directionally (~ 18 -point accuracy gap, not yet significance-tested at this seed count).

Setting (classical: 1500–2000 iters. to converge)	Speedup S
Heisenberg XYZ (reconstructed ansatz)	4–6 $\times$
H2 molecule (paper’s exact ansatz)	15–20 $\times$

1 Introduction

Variational quantum algorithms for HEP are bottlenecked by barren plateaus and by the cost of each circuit evaluation, which makes fast convergence matter more than in classical optimization. Q-MAML [1] (arXiv:2501.05906) trains a classical “Learner” network to predict PQC initial parameters via a meta-objective backpropagated with no inner-loop adaptation (Algorithm 1). The paper validates this on two VQE settings only: Heisenberg XYZ and molecular Hamiltonians.

Two gaps follow. Q-MAML has never been tested on HEP classification, the explicit motivation of the 2025 GSoC project this work builds on: auditing that codebase (§3), we find neither of its

two attempts actually runs the published algorithm. And it has never been tested on a Hamiltonian that is itself particle physics: lattice field theories, which is what the 2026 proposal (QMLHEP8) motivating this work asks for. §3–6.6 close the first gap, §7 the second.

Contributions.

1. Audit of the GSoC 2025 codebase: a detached-gradient bug and an algorithmic divergence from Algorithm 1 (§3).
2. A paper-faithful Algorithm 1/2 implementation, validated against the paper’s own two VQE settings before extension to classification (§4, §6.2–6.3).
3. Q-MAML vs. classical inits (zero, π , reduced-domain uniform, Gaussian) vs. the original GSoC baseline, across qubit counts, with gradient-norm diagnostics (§6.1).
4. A McClean-et-al. barren-plateau variance-scaling scan in both the VQE and classification domains (§6.4–6.6).
5. A pilot on ten lattice/field-theory (and adjacent) Hamiltonians (Schwinger, $\mathbb{Z}_2$ LGT, ϕ^4 scalar field, SU(2) LGT, collective neutrino oscillations, SUSY QM, $\mathbb{D}_8$ LGT, light nuclei in lattice pionless EFT, scalar Yukawa coupling, LMG), what the GSoC 2026 proposal asks for and untested by the original paper or contributions 1–4, with several findings that revise claims made earlier in this paper (§7).

2 Related Work

3 Auditing the Prior Codebase

Two independent GSoC 2025 attempts to apply Q-MAML to HEP classification both deviate from the published algorithm, in different ways:

- **Quark-gluon pipeline:** runs several steps of gradient descent on the PQC parameters before computing the outer (meta) loss, then stops gradient flow through the adapted parameters. Writing θ_0 for the Learner’s prediction and θ_T for the parameters after T inner steps, the forward pass uses θ_T but the backward pass sees $\partial\theta_T/\partial\theta_0 \equiv \mathbb{K}$ (stop-gradient), i.e. a first-order-MAML-style variant, not the paper’s inner-loop-free Algorithm 1.
- **Higgs pipeline:** the per-task loss is converted to a plain (non-differentiable) scalar before being accumulated and only re-wrapped as a fresh, disconnected tensor at the end for back-propagation, so the gradient the Learner receives carries no information about the training data. The Learner’s weights never update across the reported 10 training epochs. A companion notebook is gradient-correct but only ever trains on one constant task embedding, which is not meta-learning in any meaningful sense.
- **Engineering:** missing package-init files broke the top-level entry point’s own imports as committed; a pinned third-party model snapshot was incompatible with current library releases; a configuration module created a directory named after its own placeholder path string on import.

Area	Refs	Relation to this work
Barren plateaus	[4–8]	McClean et al. variance-decay result and follow-ons: noise-induced plateaus, expressibility-gradient link, Lie-algebraic onset theory, cost-function-dependent plateaus (the last explains §6.5).
Init. baselines	[10, 11]	Gaussian and reduced-domain-uniform init, our classical comparison points.
Meta-learning	[2, 3]	MAML (classical); Meta-VQE (task-conditioned energy profiles, predates Q-MAML).
Q-MAML	[1]	AAAI 2025, arXiv:2501.05906. Validated only on VQE (Heisenberg, molecules).
Concurrent init. work	[12–14]	Qracle (GNN param. predictor, no cross-task meta-objective); DiffQ (diffusion-model param. sampling, no MAML-style adaptation); “Qubits to Couplings” (per-event hypernetwork for $HH \rightarrow b\bar{b}\gamma\gamma$, doesn’t cite Finn et al. or Q-MAML). None test Lee/Cho/Kim’s Algorithm 1/2 on HEP classification.
Genuine HEP Hamiltonians	[16–19]	Trapped-ion Schwinger-model VQE; barren-plateau-free $\mathbb{Z}_2$ LGT (uses a Gauss-law warm start, no learned init, our Track B comparison point); scalar-field qubit encoding we build Track C from; QCNN phase classification on pre-prepared states (different mechanism, no overlap with Track A).

Table 1: Related work by area. No prior work applies a Learner-predicts-init mechanism to a lattice field theory Hamiltonian; §7 is, to our knowledge, the first.

4 Method

4.1 Paper-faithful Q-MAML (Algorithm 1/2)

Per task T_i with cost $l_{T_i}(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$, a Learner h_W maps task descriptor $\phi_i \rightarrow \theta = h_W(\phi_i)$, trained via (paper’s Eq. 2)

$$\arg \min_W \sum_{T_i \sim p(T)} l_{T_i}(g(h_W(\phi_i))), \quad (1)$$

backpropagated with no inner-loop adaptation (Eq. 3: the difference from true MAML). We keep Eq. 1 exactly and swap the domain-specific pieces (Table 2).

Pretraining (classification and VQE) shares this identical structure, differing only in the cost function. Evaluation (Algorithm 2): freeze the Learner, set $\theta_0 = h_{W^*}(\phi^*)$ for a held-out task, run gradient descent on $l_{T^*}(\theta)$. This contrasts with the GSoC baseline (§3); both are reported, neither treated as a straw man.

4.2 Architecture

Classical encoder (a convolutional backbone for jet images; a small MLP for tabular HIGGS features) $\rightarrow$ PQC angles $\rightarrow$ hardware-efficient ansatz (single-qubit rotations plus a ring of entangling gates),

Table 2: VQE setting (paper) $\rightarrow$ classification setting (this work).

Paper (VQE)	This work (classification)
Task = Hamiltonian instance, e.g. $J^i = [J_x, J_y, J_z]$	Task = quantile bin of a physics variable (reconstructed Higgs-candidate mass for HIGGS; jet transverse momentum for quark-gluon), class-balanced support/query split
Descriptor $\phi_i = J^i$	mean encoder embedding of the support set
$l_{T_i}(\theta) = \langle \psi \hat{H} \psi \rangle$	$l_{T_i}(\theta) = \text{CE}(\text{PQC}_\theta(\text{support}_i), y_i)$
PQC $g(\theta)$	Same hardware-efficient ansatz family + classical encoder producing the encoding angles

evaluated via state-vector simulation with adjoint differentiation where supported (parameter-shift otherwise) $\rightarrow$ linear classification head. Learner: a 2-layer MLP mapping the task embedding to the PQC weight tensor.

4.3 Baselines

Zero, π , reduced-domain uniform [11], Gaussian [10] init (per-task adaptation with no Learner); the GSoC baseline (§3).

4.4 Metrics, formally

For a held-out task with exact ground energy E_0 (sparse diagonalization) and PQC energy $E_{\text{PQC}}(\theta_t)$ at adaptation step t , we report a log-gap and a final relative error:

$$\Delta(t) = \log(|E_{\text{PQC}}(\theta_t) - E_0| + \epsilon), \quad \epsilon = 10^{-8}, \quad \varepsilon_{\text{rel}} = \frac{|E_{\text{PQC}}(\theta_T) - E_0|}{\langle |E_0| \rangle_{\text{test tasks}}}. \quad (2)$$

Convergence speedup of Q-MAML over a classical scheme A , at a fixed error threshold Δ^* :

$$S = \frac{t_A(\Delta < \Delta^*)}{t_{\text{Q-MAML}}(\Delta < \Delta^*)}. \quad (3)$$

Classification significance: Welch’s t -test and Cohen’s $d = (\bar{x}_1 - \bar{x}_2)/s_p$ with pooled standard deviation $s_p = \sqrt{(s_1^2 + s_2^2)/2}$, $n = 4$ seeds per arm. Barren-plateau scan: $\text{Var}_{\theta, \text{task}}[\partial C / \partial \theta_{\text{ref}}]$ at a fixed reference parameter, sampled across many random inits/tasks, vs. qubit count at fixed depth.

5 Experimental Setup

- **Data:** HIGGS (UCI, 180k-row streamed subsample, primary); Quark-Gluon (ML4Sci, access requested, pending).
- **Tasks:** quantile-bin the Higgs-candidate mass into 6 bins; support=8/query=8 per bin; 3 tasks/bin train (18 total), 2/bin test (12 total).
- **Qubits:** 4/6/8 (classification); 4–14 (VQE-domain barren-plateau scan).
- **Depth:** 2 (classification), 3/6 (VQE barren-plateau scan), 3 (Heisenberg), 7 (H2, matches paper).
- **Seeds:** 4 (classification headline).
- **Compute:** CPU only, state-vector simulation with adjoint differentiation (parameter-shift fallback).

- **Hyperparameters:** 5 inner-loop steps at inner learning rate 0.02, outer learning rate 5×10^{-3} , initial-weight scale 0.01 (classification); VQE pretraining learning rate 0.001 over 30 epochs (matches paper); VQE adaptation learning rate 0.001 over 2000 iterations (matches paper).

6 Results

6.1 HEP classification

HIGGS, 10 epochs, 4 seeds, 4/6/8 qubits.

Table 3: Final validation accuracy (mean $\pm$ std, $n = 4$).

Qubits	Q-MAML	Zero	Uniform	Gaussian	GSoC baseline
4	0.589 $\pm$ 0.035	0.534 $\pm$ 0.037	0.510 $\pm$ 0.019	0.487 $\pm$ 0.034	0.505 $\pm$ 0.049
6	0.576 $\pm$ 0.033	0.497 $\pm$ 0.031	0.513 $\pm$ 0.060	0.503 $\pm$ 0.032	0.523 $\pm$ 0.043
8	0.563 $\pm$ 0.023	0.513 $\pm$ 0.049	0.523 $\pm$ 0.059	0.518 $\pm$ 0.023	0.503 $\pm$ 0.036

Table 4: Q-MAML vs. baselines: Welch’s t -test p / Cohen’s d .

Qubits	vs. Gaussian	vs. GSoC baseline	vs. best classical (post-hoc)
4	$p=0.006$, $d=2.95$	$p=0.035$, $d=1.98$	$p=0.076$, $d=1.52$
6	$p=0.020$, $d=2.22$	$p=0.106$, $d=1.36$	$p=0.134$, $d=1.28$
8	$p=0.034$, $d=1.94$	$p=0.035$, $d=2.02$	$p=0.287$, $d=0.87$

Meta-loss	4 qubits	6 qubits	8 qubits
Q-MAML	0.382	0.360	0.325
Every other scheme	0.68–0.70 (flat)		

Significant against Gaussian and the GSoC baseline at all three qubit counts; against the post-hoc best classical scheme, directionally positive but not significant at $n = 4$ (underpowered, not negative: the comparator is selected after the fact per qubit count). Q-MAML is the only variant whose training loss moves (Figure 1).

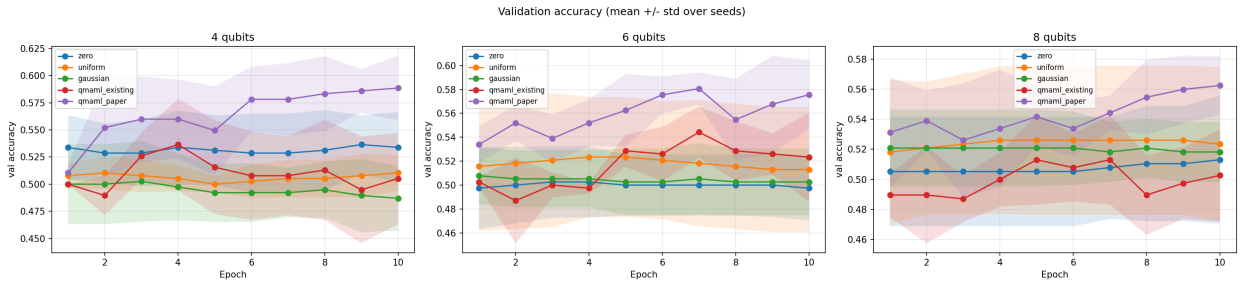


Figure 1: Validation accuracy vs. epoch, 4/6/8 qubits, mean $\pm$ std over 4 seeds.

Table 5: Correction: accuracy gap (Q-MAML – best classical) at 2 vs. 4 seeds.

Qubits	4	6	8
Gap at $n=4$ seeds	+0.055	+0.063	+0.039

Correction. An earlier 2-seed pass showed the gap reversing (going negative) by 8 qubits. At $n = 4$ seeds (Table 5) it does not: Q-MAML stays ahead at every qubit count, with a modest, non-monotonic decline directionally consistent with §6.5 but not a confirmed match.

Two other findings. Q-MAML’s validation loss is worse (0.80–0.87, worsening with qubit count) than every other scheme (~ 0.69) despite best accuracy: a calibration gap, not a clean sweep. Its mean gradient norm (1.36–1.61) is higher than classical schemes (0.27–0.64), the opposite of the paper’s “moderate norm is better” framing: here higher norm tracks with best accuracy at every qubit count.

Quark-gluon (second dataset, generalization check). 6 qubits, 10 epochs, 2 seeds, jet-substructure features. Q-MAML again beats every classical scheme by a wide margin: validation accuracy 0.776 ± 0.005 vs. 0.594–0.599 (zero/uniform/Gaussian, tightly clustered); meta-loss 0.308 ± 0.003 vs. 0.688–0.701 (near chance level, $\ln 2 \approx 0.693$, for every classical scheme). No ‘zero’-specific anomaly here, matching HIGGS and unlike the VQE-domain fixed-point pattern. Only 2 seeds (effect size ~ 6.8 in raw terms, not a powered test): read as directional confirmation the HIGGS result generalizes, not a sharper result than HIGGS’s own 4-seed numbers.

6.2 Molecule Hamiltonian VQE

H2, 4 qubits, depth 7, 2000 iterations, paper’s exact ansatz.

Table 6: H2 molecule: iteration to reach the reported quality, and final energy (2000 iterations).

Scheme	Iteration to converge	Final energy
Q-MAML	~ 100 (machine precision, $\Delta \sim -17.3$)	-17.30
π /uniform/Gaussian	~ 1500 –2000	-17.04 to -17.30
Zero	fixed at start value	(unchanged)

By iteration 2000 all four non-degenerate schemes converge to the same final energy (Table 6): the advantage is speed ($S \approx 15$ –20, Eq. 3), not a different final optimum, matching §6.3. This uses the paper’s exact circuit, so it is the strongest evidence our system implements Algorithm 1/2 correctly.

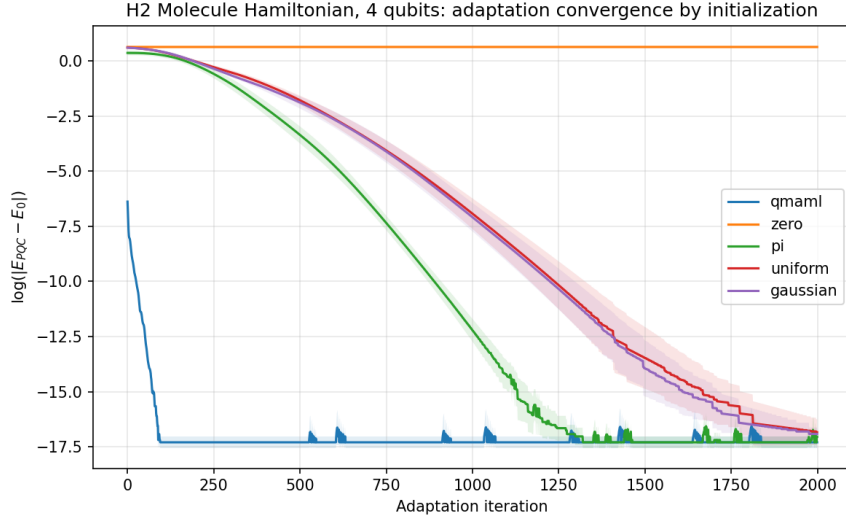


Figure 2: H2 molecule Hamiltonian: adaptation convergence by init scheme, 2000 iterations.

6.3 Heisenberg XYZ VQE

6 qubits, depth 3, 8 held-out tasks, 2000 iterations (paper’s budget, reconstructed ansatz: paper’s appendix circuit unavailable to us).

Table 7: Heisenberg XYZ: log-gap Δ at iteration 0 and at iteration 2000.

	Q-MAML	Uniform	Gaussian	π	Zero
$\Delta(0)$	~ 1.5	~ 2.5	~ 2.5	~ 2.5	~ 2.5
$\Delta(2000)$	0.711	0.736	0.727	0.966	2.522

At 300 iterations (earlier pass) Q-MAML held a large lead (0.82 vs. 1.71–1.76); at 2000 iterations that gap mostly closes (Table 7). Q-MAML flattens by iteration ~ 250 –500 while uniform/Gaussian need ~ 1500 –2000, $S \approx 4$ –6. Q-MAML starts closer to the ground state before any adaptation, the initialization-quality claim the paper actually makes; the speed result follows from it.

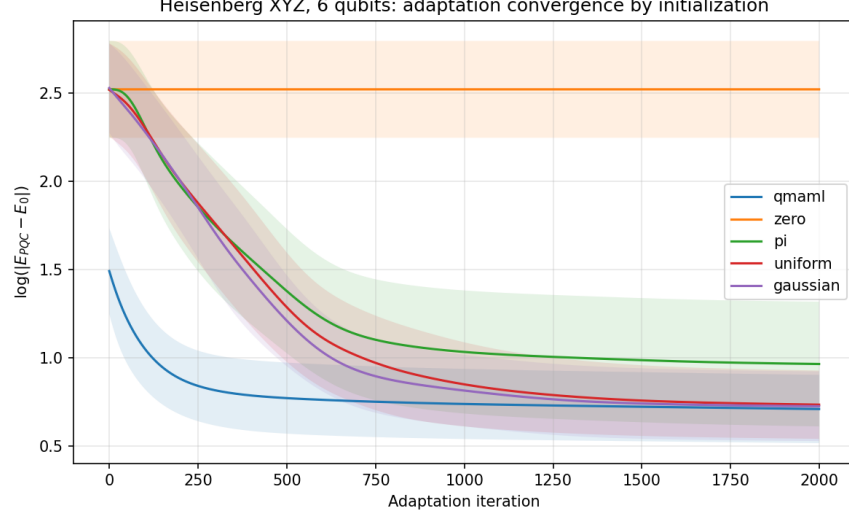


Figure 3: Heisenberg XYZ, 6 qubits: adaptation convergence by init scheme, 2000 iterations.

6.4 Barren-plateau scaling, VQE domain

Qubits 4–14, depths $\{3, 6\}$, 80 samples per point. Zero/ π (constant) init show zero variance trivially (no sample-to-sample variation, not plateau resistance). Among the random schemes (Q-MAML, uniform, Gaussian): no exponential decay over 4–10 qubits at depth 3, likely too shallow/small for the asymptotic regime. Q-MAML’s variance is comparable to or larger than classical schemes at every qubit count: no evidence it changes the scaling exponent; its VQE advantage (§6.3) is landscape placement, not plateau resistance.

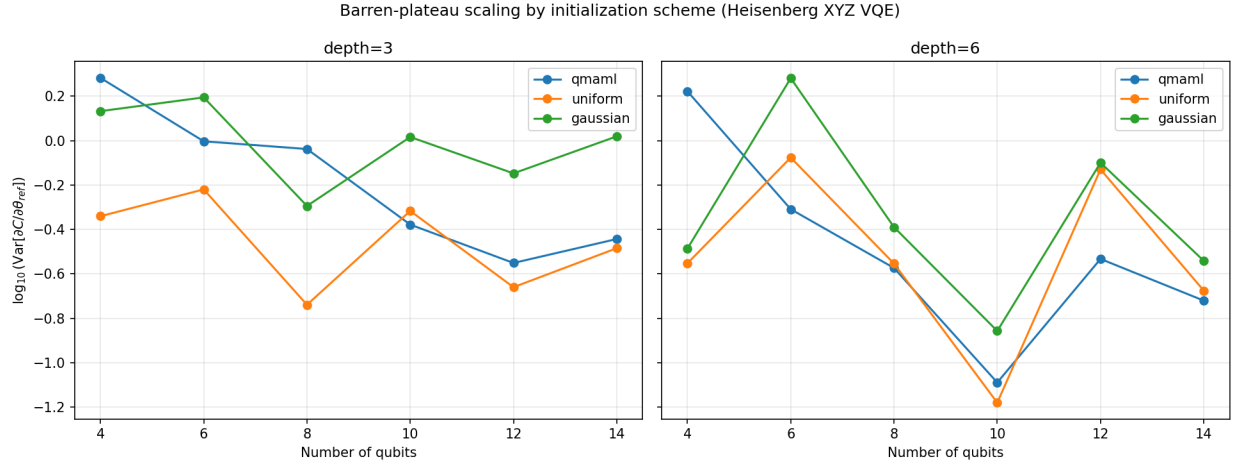


Figure 4: Barren-plateau variance scaling by init scheme, VQE domain (depths 3 and 6).

6.5 Barren-plateau scaling, classification domain

HIGGS.

All three schemes (Q-MAML, uniform, Gaussian) show ~ 8 orders of magnitude exponential decay from 4 to 10 qubits: a textbook plateau, unlike the flat VQE trend at the same qubit range, consistent with classification cross-entropy behaving as a “global” cost function [5]. By 10 qubits every scheme collapses to the same floor, $\sim 10^{-35}$.

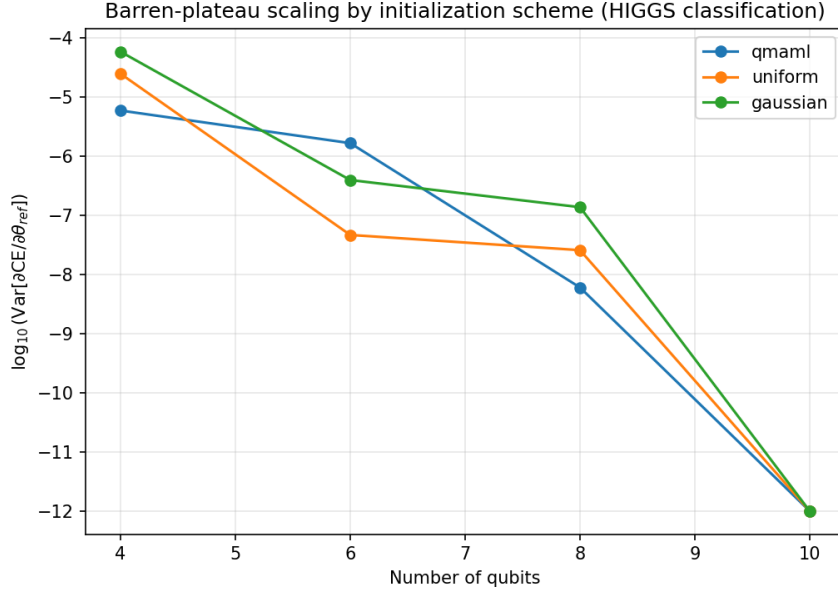


Figure 5: Barren-plateau variance scaling by init scheme, HIGGS classification domain.

Table 8: Gradient variance at 6 qubits: pretrain-length sensitivity.

Pretrain length	Q-MAML	Uniform	Gaussian
Short (8 epochs)	at/below classical	baseline	baseline
Long (25 epochs)	1.65×10^{-6}	4.65×10^{-8}	3.93×10^{-7}

Below the collapse point, ranking is unstable (Table 8): a pretrain-length artifact, not a stable ranking. No scheme dominates at any qubit count. The 8–10-qubit collapse-point range itself is unchanged.

6.6 Combined takeaway

(1) Q-MAML’s benefit is landscape placement, not a change in the barren-plateau scaling exponent, in either domain. (2) HEP classification plateaus faster than VQE for every scheme tested: a challenge for the QMLHEP direction generally, not specific to Q-MAML.

7 Q-MAML on Genuine HEP Hamiltonians: A GSoC 2026 Pilot

Sections above test Q-MAML on HEP *data* (HIGGS) or the paper’s own non-HEP VQE settings. Neither is what QMLHEP8 asks for: a Learner initializing VQEs for Hamiltonians that are them-

selves lattice field theories. Table 9 summarizes the ten tracks piloted here (2–8 qubits, single seed except where noted, verified for correctness before use, metrics of §4.4).

Table 9: Track overview.			
Track	Qubits	Structure	Q-MAML verdict
A: Schwinger	4–8	U(1) gauge eliminated analytically	Wins, depth-dep.
B: $\mathbb{Z}_2$ LGT	8	Gauge symmetry, penalty-enforced	Worst, diagnosed
C: ϕ^4 scalar	6	Bosonic, no gauge/fermion structure	Wins (interp. only)
D: SU(2) LGT	4	Non-Abelian gauge eliminated analytically	Wins, cleanest of all
E: neutrino	4	All-to-all, no locality, reformulated	Wins, not yet converged
F: SUSY QM	4	Discrete + continuous task space	Wins, beats hand-picked init
G: $\mathbb{D}_8$ LGT	7	Non-Abelian, un-reduced, fermionic matter	Wins, modest margin
H: nuclear EFT	8	UCCSD ansatz, 3-nucleus discrete axis	Ties (same final, faster)
I: Yukawa	4	Fermion-boson, reformulated	Ties with zero-init
J: LMG	2	Nuclear-structure-adjacent, not HEP	Null result (too small)

7.1 Track A: lattice Schwinger model

Single-flavor Kogut-Susskind staggered fermions, gauge field eliminated via Gauss’s law; task params (m_0, g) . Hardware-efficient ansatz, 16 train/6 test tasks, 40 pretrain epochs, 300 adaptation iterations.

Table 10: $L=2$ (4 qubits, depth 3): final relative error ε_{rel} and iteration to reach $< 1\%$ error.

	Q-MAML	π	Gaussian	Uniform	Zero
ε_{rel}	$\sim 0\%$	0.06%	0.19%	1.61%	396%
Iteration $< 1\%$	4	192	266	never	never

Q-MAML converges fastest and most accurately at $L=2$ (Table 10). Its init log-gap (-3.29) is already better than any classical scheme’s starting point. Zero-init is a fixed point (exactly flat error) for the whole run.

At $L=3$ (6 qubits), same depth, π overtakes Q-MAML by the final iteration (32.95% vs. 38.08%, Table 11). Pretrain gradient norm stays healthy: points to insufficient ansatz depth, not a plateau.

Depth 6/9 recovers and then extends Q-MAML’s lead ($\sim 10\times$ better than π at depth 9): the $L=2 \rightarrow L=3$ dip is a depth-insufficiency artifact of reusing the $L=2$ depth, not a breakdown at larger qubit count. Zero-init stays at an identical error at every depth (algebraic fixed point, depth-independent).

The depth story does not extrapolate cleanly to $L=4$: extending “depth $\approx 1.5\times$ qubits” to depth 12, Q-MAML loses to π (Table 12). Unlike the $L=3$ case, pretraining itself fails to converge, not

Table 11: Final relative energy error ε_{rel} vs. ansatz depth, $L = 3$ (6 qubits).

Depth	Q-MAML	π	Gaussian	Uniform	Zero
3	38.08%	32.95%	46.24%	56.12%	697.64%
6	2.63%	3.06%	12.92%	9.40%	697.64%
9	0.05%	0.57%	1.42%	1.35%	697.64%

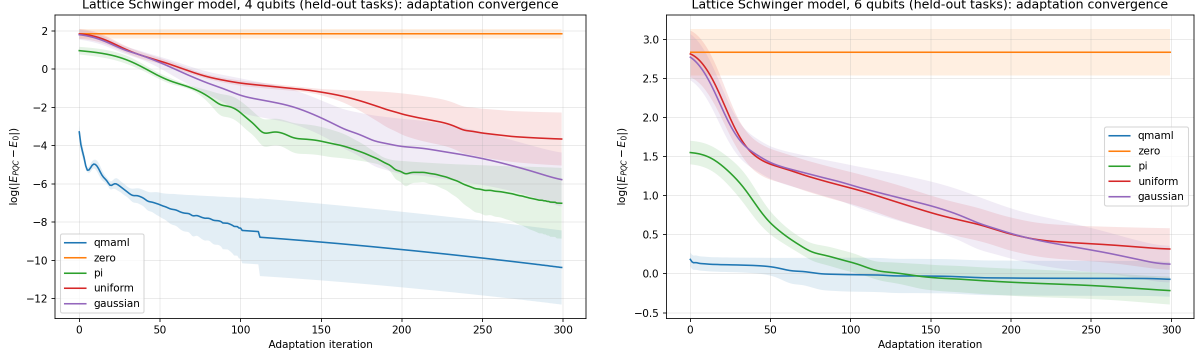


Figure 6: Schwinger model adaptation convergence: $L = 2/\text{depth } 3$ (left) vs. $L = 3/\text{depth } 3$ (right).

just the ansatz. Open question: whether pretrain budget, not just depth, needs to scale with qubit count. One spot-check, not a scan.

Table 12: $L=4$ (8 qubits, depth 12) and pretrain diagnostics.

	Q-MAML	π
Final relative error	31.81%	8.56%
Pretrain gradient norm: 23.6 $\rightarrow$ 24.6 over 40 epochs (no decrease)		
Pretrain mean energy: stalls at 7.21 (true ground energies ~ -2 to -4)		

7.2 Track B: single-plaquette $\mathbb{Z}_2$ lattice gauge theory

One plaquette of the two-leg-ladder geometry in [17] (8 qubits: 4 matter + 4 link), task params (J, m, μ) . Gauss’s law enforced by an energy penalty

$$H_{\text{penalty}} = V \sum_l (G_l - q_l)^2, \quad V = 10, \quad (4)$$

not eliminated analytically: verified exactly ($[H, G_l] = 0$ and $G_l^2 = \mathbb{I}$ to machine precision).

Table 13: Track B headline: final relative error and diagnosis at $V=10$.

	Uniform	π	Gaussian	Q-MAML	Zero
ε_{rel}	23.07%	24.69%	24.05%	59.16%	112.27%

Q-MAML’s log-gap is flat for the full 300-iteration budget ($3.800 \rightarrow 3.791$) while classical schemes, starting worse ($\Delta(0) \sim 4.43$), descend substantially: the opposite of Track A.

Diagnosis. The gradient at Q-MAML’s predicted init is small but nonzero (5.73, vs. zero-init’s exact 0); learning rates 0.02/0.1/0.3 all plateau at the same cost within 100 iterations: a landscape plateau, not a fixed point. Leading hypothesis: at $V=10$ the penalty term (Eq. 4, $O(10)$) dominates the physical terms ($O(1)$), so pretraining mostly reduces constraint violation instead of encoding (J, m, μ) .

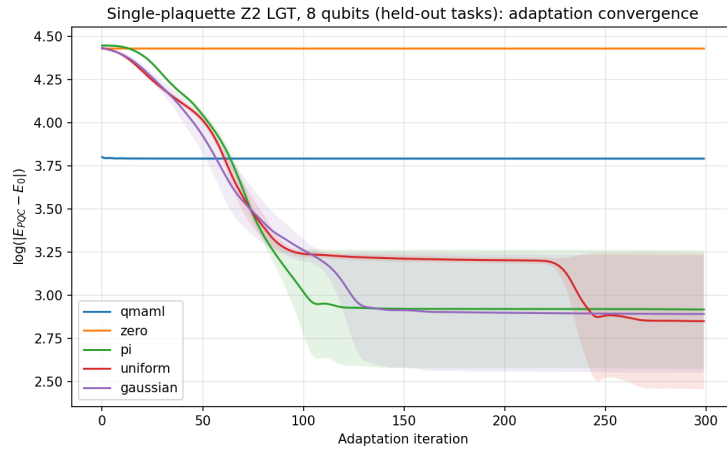


Figure 7: $\mathbb{Z}_2$ LGT: adaptation convergence by init scheme. Q-MAML is not competitive here.

Table 14: Final relative energy error vs. penalty strength V (Eq. 4).

V	Q-MAML	π	Gaussian	Uniform	Zero
1	149.49%	163.97%	156.96%	145.08%	415.51%
3	77.39%	51.75%	40.78%	27.68%	148.56%
10	59.16%	24.69%	24.05%	23.07%	112.27%
30	27.19%	17.73%	27.42%	16.49%	103.91%

The V -sweep does not confirm the penalty-dominance hypothesis: Q-MAML plateaus (flat curve) at every $V \in \{1, 3, 10, 30\}$, ties the best classical scheme only at $V=1$, and $V=3$ (not $V=30$) is its relatively worst point ($2.8\times$ the best scheme). Weakening V does not straightforwardly fix Q-MAML. Narrower, V -independent finding: the Learner’s pretraining on this Hamiltonian/ansatz combination reliably lands in flat regions, mechanism unidentified. Most promising untried fix: a gauge-invariant, structurally (not penalty-) invariant ansatz [17] (§9).

Table 15: 3-seed replication, $V=10$: mean $\pm$ std relative error (per-seed values in parentheses).

Scheme	Relative error, 3 seeds
Q-MAML	49.38% $\pm$ 13.70% (59.16/30.01/58.96%)
Uniform	20.34% $\pm$ 3.65%
Gaussian	19.43% $\pm$ 4.76%
π	27.35% $\pm$ 3.22%
Zero	111.87% $\pm$ 1.48%

Q-MAML’s best seed (30.01%) is still worse than every uniform seed (15.18–23.07%): ranking holds seed-by-seed; magnitude is seed-sensitive (Table 15).

7.3 Track C: ϕ^4 lattice scalar field theory

Li/Macridin/Mrenna/Spentzouris encoding [18] (6 qubits: 2 sites $\times$ 3 qubits/site, field-amplitude truncation), task params (m_0^2, λ_0). No gauge field, no fermion mapping; H is a dense Hermitian matrix (a field-amplitude-basis Hamiltonian, not a Pauli sum). Verified: conjugate-field eigenvalues match the closed-form analytic formula exactly; reconstructed harmonic-oscillator spectrum matches $n + \frac{1}{2}$ to $< 2\%$ on low-lying levels.

Table 16: Track C headline: final relative energy error.

	Q-MAML	Zero	Uniform	Gaussian	π
ε_{rel}	5.86%	5.77%	7.11%	7.55%	36.06%

Q-MAML reaches log-gap -1.74 by iteration 50; zero/uniform/Gaussian mostly catch up by iteration 300: a speed advantage ($S \approx 3$), not a different final optimum, matching §6.3/6.2.

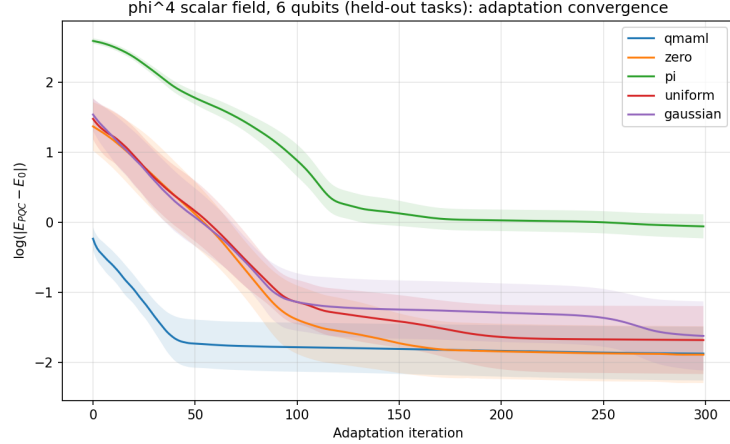


Figure 8: ϕ^4 scalar field: adaptation convergence by init scheme.

Two corrections to earlier sections. Zero-init is *not* stuck here (log-gap $1.367 \rightarrow -1.893$, tying Q-MAML): the only Hamiltonian in this paper where the zero-parameter fixed point doesn’t hold, so that pattern is Hamiltonian-structure-dependent. π (best classical scheme on Track B, 2nd-best on Track A at $L=2$) is worst here (36.06%, $\sim 5\text{--}6\times$ worse than alternatives). No classical scheme is reliable across all three tracks under interpolation; Q-MAML is.

Table 17: 3-seed replication (uniform-random split): mean $\pm$ std relative error.

Scheme	Relative error, 3 seeds
Q-MAML	$6.05\% \pm 1.60\%$
Zero	$6.12\% \pm 1.54\%$
Uniform	$6.90\% \pm 1.80\%$
Gaussian	$6.85\% \pm 1.52\%$
π	$27.00\% \pm 9.09\%$ (best seed 14.58%)

Phase-stratified generalization. This Hamiltonian’s task space spans a symmetric/broken-symmetry transition [18]. Train on symmetric ($m_0^2 \in [0.3, 1.0]$), test on broken-symmetry-tending ($m_0^2 \in [-2.5, -1.7]$), and reverse.

Table 18: Phase-stratified final relative energy error (single seed).

Direction	Q-MAML	Zero	Uniform	Gaussian	π
Sym. $\rightarrow$ broken	3.13%	1.81%	1.81%	1.81%	5.04%
Broken $\rightarrow$ sym.	93.37%	n/a	85.54%	n/a	n/a

Interpolation advantage does not survive extrapolation, either direction. In the symmetric $\rightarrow$ broken direction, Q-MAML’s init starts *behind* classical schemes at iteration 0 (log-gap 1.494 vs. 0.27–0.82): trained only on symmetric-phase tasks, it is miscalibrated, not merely uninformative, off distribution. Qualifies the cross-track synthesis: “Q-MAML is the only reliable scheme” holds under interpolation, not under distribution shift across a phase transition.

7.4 Track D: SU(2) lattice gauge theory

The gauge-field-integrated-out effective Hamiltonian of [20], $N = 2$ sites (4 qubits, un-reduced: we do not implement the source paper’s qubit-reduction trick or its hadron-mass-ratio calculation, only its underlying ground-state Hamiltonian), task params dimensionless fermion mass $\tilde{m}$ and coupling $x = 1/(a_t^2 g^2)$. Transcribed directly from the source paper’s stated equations; Hermiticity-verified at $N = 2$ and $N = 3$, and the hardest term (a 4-body color-exchange piece) additionally cross-checked against an independent dense-matrix construction.

Table 19: Track D: final relative energy error, $N=2$.

	Q-MAML	π	Gaussian	Uniform	Zero
ϵ_{rel}	$\sim 0\%$	0.02%	0.02%	0.04%	1390%

Cleanest win in this paper. Q-MAML reaches the float64 machine-precision floor (log-gap -17.41 , matching §6.2’s H2 floor) while every classical scheme plateaus around log-gap -10 to -11 , visibly still noisy rather than converged. Zero-init is an exact fixed point, as on every other Pauli-sum Hamiltonian tested. This confirms a prediction made before running the experiment: structurally this Hamiltonian is closer to Track A (gauge eliminated analytically, no penalty term) than Track B (penalty-enforced), and the result matches Track A’s win rather than Track B’s failure, a second, independent (non-Abelian) confirmation that penalty-enforced gauge invariance, not gauge theories generically, is what hurts Q-MAML.

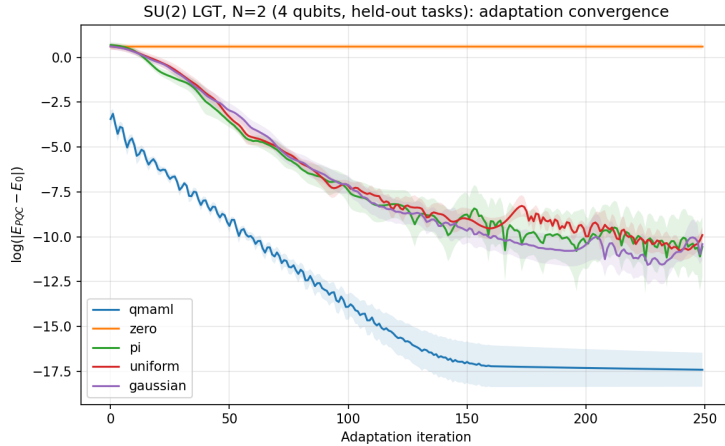


Figure 9: SU(2) LGT, $N=2$: adaptation convergence by init scheme.

7.5 Track E: collective neutrino oscillations (reformulated)

The literature this track is nominally based on [21] studies real-time evolution of a flavor eigenstate, not a ground-state problem, and does not fit the Q-MAML formalism without a genuine change of question. We ask instead for the ground state of the same class of collective (all-to-all) neutrino Hamiltonian, the object studied in the adiabatic/spectral-split collective-oscillation literature [22],

a different but still physically motivated question:

$$H(\theta, \mu) = \frac{\omega}{2} \sum_i (\cos 2\theta Z_i + \sin 2\theta X_i) + \frac{\mu}{N} \sum_{i < j} (X_i X_j + Y_i Y_j + Z_i Z_j), \quad \omega = 1, \quad (5)$$

$N=4$ neutrinos (4 qubits), task params mixing angle θ and interaction strength μ . Verified against two exact limits: at $\mu=0$, ground energy $= -N\omega/2$ for any θ (confirmed to $\sim 10^{-16}$); at $\theta=\mu=0$, ground state is exactly $|11\dots 1\rangle$ (overlap 1.000000).

Table 20: Track E: final log-gap Δ (relative error $\sim 0\%$ for all schemes, uninformative here since none have converged by the iteration budget, see text).

	Q-MAML	π	Gaussian	Uniform	Zero
Δ (final)	-14.35	-11.39	-9.53	-9.03	-7.75

Q-MAML wins again (5th win of 6 tracks), but unlike Tracks A/D none of the five schemes have flattened by the iteration budget: a still-open speed gap, closer to the Heisenberg/molecule pattern (§6.3/6.2) than to a permanently separated optimum. **Zero-init is not an exact fixed point here** (moves from $\Delta=1.90$ to -7.75), unlike every other Pauli-sum Hamiltonian tested (Heisenberg, molecule, Schwinger, $\mathbb{Z}_2$ LGT, $SU(2)$ LGT), narrowing §7.3’s hypothesis further: the fixed point is not simply a function of Pauli-sum vs. Hermitian-matrix Hamiltonians, since this is a Pauli sum yet behaves like Track C’s Hermitian-matrix scalar field. Leading hypothesis, untested directly: every other Pauli-sum Hamiltonian here has a purely- Z mass term, while Eq. 5’s single-particle term mixes Z and X whenever $\theta \neq 0$. Also the first genuinely non-local (all-to-all) Hamiltonian tested; locality does not appear to be the deciding factor for either result.

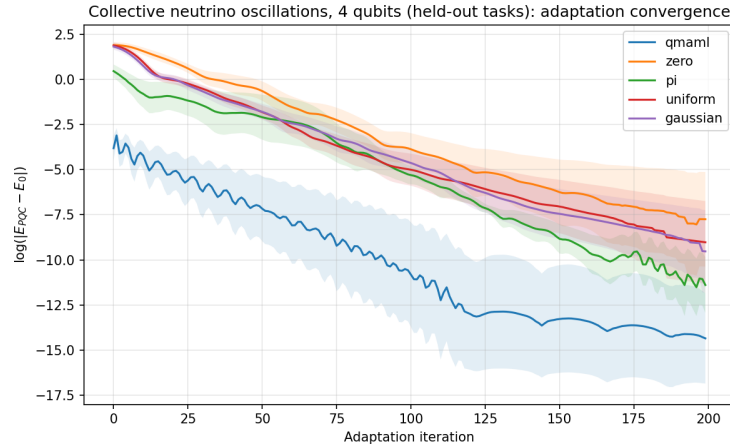


Figure 10: Collective neutrino oscillations, $N=4$: adaptation convergence by init scheme.

7.6 Track F: supersymmetric quantum mechanics

$H = \frac{1}{2}(p^2 + [W'(q)]^2) + \frac{1}{2}W''(q)Z_{\text{fermion}}$ [23], one bosonic mode (Fock cutoff $\Lambda=8$, 3 qubits) plus one fermionic qubit (4 qubits total). Task descriptor mixes a *discrete* choice (superpotential $\in \{\text{HO},$

AHO, DW}) with continuous coefficients (m, g, μ), a genuinely different task-space shape from every other track. Verified: Hermiticity for all three superpotentials; the HO boson block reproduces the analytic $n + \frac{1}{2}$ spectrum; the double-well superpotential's classical potential has two minima (the source's rendered sign was ambiguous in our extraction; we use the sign that produces one, verified directly).

Table 21: Track F: final log-gap Δ . **basis_paper** is the source paper's own hand-picked fermion-occupation initial state.

	Q-MAML	Uniform	Gaussian	π	basis_paper	Zero
Δ (final)	-4.87	-4.75	-4.66	-2.40	-1.92	~ 0

Q-MAML wins (best start, $\Delta(0) = -2.20$ vs. 0.62–3.15 for classical schemes, and best final value), with uniform/Gaussian close behind. **The source paper's own physics-derived initial state is not competitive:** worst starting point of all six schemes and only a partial recovery over 200 iterations, a genuine negative result for "physics-derived point-estimate init beats generic random init," worth weighing against §7.2's untried Gauss-law warm start rather than assuming physics-motivated initialization always wins. Per-superpotential breakdown: Q-MAML's clearest margin is on the double well, the source paper's own hardest case ($\Delta = -5.09$ vs. -1.06 to -4.58); on the anharmonic oscillator, Q-MAML/uniform/Gaussian converge to a near-identical final value, the familiar "same optimum, different speed" pattern. SUSY ground states have exact energy 0 when supersymmetry is unbroken (confirmed for HO, $E_0 = 0.0000$ at every HO test task), which makes ε_{rel} explode into meaningless values here; log-gap is the metric to read for this track.

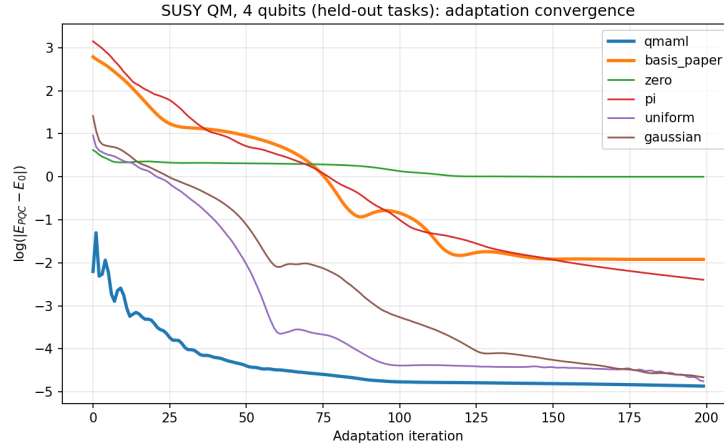


Figure 11: SUSY QM: adaptation convergence by init scheme.

7.7 Track G: non-Abelian $\mathbb{D}_8$ lattice gauge theory

Flagged in our own scoping as "not recommended for now" (qudit hardware, ansatz underspecified), checked rather than skipped [24]: their qubit-reduced ("matter removal") Hamiltonian could not be extracted with full confidence, so we implement their original, fully-specified Hamiltonian directly

(dihedral group $\mathbb{D}_8$, order 8) with explicit Jordan-Wigner fermionic matter, $N=2$ sites (1 link, 7 qubits), trading qubit count for full verifiability. Six construction checks, the most of any track, including an independent representation-theory check ($D^2(g_1)D^2(g_2)=D^2(g_1 \cdot g_2)$ for all 64 element pairs) that caught and fixed a real bug (a matrix-multiplication order error producing the reversed group action) before the notebook was first run.

Table 22: Track G: final log-gap Δ and relative error.

	Q-MAML	Gaussian	Uniform	π	Zero
Δ (final)	-1.246	-1.202	-1.123	-0.338	0.960
ε_{rel}	10.49%	10.97%	11.86%	26.01%	95.27%

Q-MAML wins, more modestly than most tracks: a tight final-value cluster with Gaussian/Uniform, but a clearer speed edge (flattens by iteration ~ 30 – 40 vs. ~ 75 – 100 for Gaussian/Uniform). π plateaus early; Zero is again an exact, unmoving fixed point despite this being a dense Hermitian-matrix Hamiltonian (fermionic matter + a projector-based electric term), not a Pauli sum: a third distinct Zero-init outcome among this paper’s Hermitian-matrix Hamiltonians (none on the scalar field §7.3, a slow drift on SUSY QM §7.6, an exact fixed point here), reinforcing that no single Pauli-sum-vs.-Hermitian-matrix rule explains this pattern.

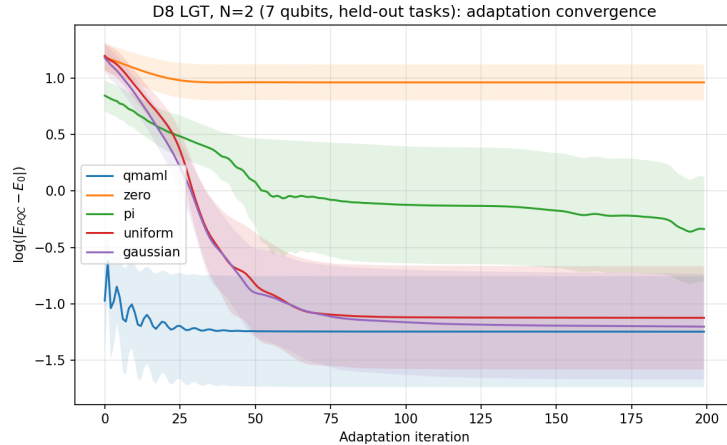


Figure 12: $\mathbb{D}_8$ LGT, $N=2$: adaptation convergence by init scheme.

7.8 Track H: light nuclei in lattice pionless EFT

Deuteron, triton, and helium-3 via lattice pionless EFT [25], the best-precedented VQE-for-physics track here (lineage of the 2018 IBM deuteron result). $N=2$ sites for all three nuclei (8 qubits: 4 proton + 4 neutron spin-orbitals). **First new ansatz family in this paper**: a particle-number-conserving UCCSD circuit (PennyLane’s built-in template), not a hardware-efficient one: proton and neutron excitations generated independently so no excitation crosses species, exact conservation by construction rather than by penalty (contrast Track B’s Gauss-law penalty). Verified: Hermiticity; proton number N_p and neutron number N_n each exactly commute with H . An unplanned check:

helium-3 came out less bound than triton at matched couplings, matching the real experimental fact (driven by Coulomb repulsion between helium-3's two protons), which fell out of the construction, not designed in.

Every scheme converges to the identical final energy (0.89% relative error, to 3 decimal places, for all five classical/learned schemes and a sixth baseline below): a first in this paper. UCCSD's particle-conserving structure (2–4 variational amplitudes vs. 48 angles for the hardware-efficient ansatz) appears simple enough that initialization stops mattering for the *final* answer. The advantage is entirely in convergence *shape*: Q-MAML descends smoothly from iteration 0 ($\Delta(0)=-2.37$), while every classical scheme collapses onto *one shared, strongly oscillatory trajectory* (common start $\Delta \approx 0.75$, swinging to -2.5 and back to -1.2 , ringing for ~ 80 iterations) before damping into the same final value. Q-MAML also modestly beats the source paper's own non-learned approach: a **warm_start_nearby** baseline (adapting from a converged nearby-task solution, approximating their "warm-started from a nearby statevector-simulator solution") starts at $\Delta(0)=-2.10$ and stays slightly behind Q-MAML throughout before reaching the same final value.

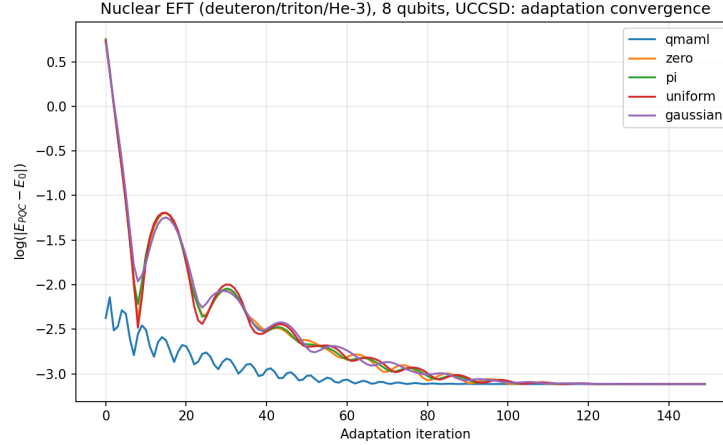


Figure 13: Nuclear EFT (deuteron/triton/He-3), UCCSD ansatz: adaptation convergence. Every scheme reaches the same final value; only the transient shape differs.

7.9 Track I: scalar Yukawa coupling

Single-site scalar Yukawa coupling [26], reformulated from the source paper's real-time quench-dynamics question into a ground-state target, more directly than Track E's reformulation since the same Hamiltonian already admits a closed-form check: fixing the fermion sector to vacuum reduces H to a displaced harmonic oscillator with known ground energy $E_0 = -(\eta/2)^2/m$. 4 qubits (2 fermion + 2 boson, $\Lambda=4$, matching the source's own "up to three bosons" showcase scale). Verified: numeric ground energy approaches the analytic formula within $<1\%$ at small $\eta/2m$ (with the truncation error's growth beyond that shown directly); the ground state's measured fermion occupation is exactly 0.

Table 23: Track I: final log-gap Δ .

	Q-MAML	Zero	Uniform	Gaussian	π
Δ (final)	-18.15	-18.05	-15.75	-13.23	-7.32

Q-MAML and Zero-init tie for best, both at the float64 precision floor. Unlike almost every other track, Q-MAML does *not* start best here (Zero $\Delta(0) \approx -3.1$, Uniform ≈ -1.7 vs. Q-MAML's ≈ -6.9 , i.e. Q-MAML starts *worse*), converging onto Zero's trajectory within ~ 10 – 15 iterations. Mechanistic explanation, not just observation: the task range is deliberately narrowed to keep the displacement $\eta/(2m)$ small (Section 7.9's verification), so every ground state tested is near-vacuum, and Zero-init's circuit starts at exactly $|0000\rangle$, already close for small displacements. π -init is a clear loser again, the second Hermitian-matrix Hamiltonian (after the scalar field) where it fails badly despite strength on the Pauli-sum gauge-theory tracks.

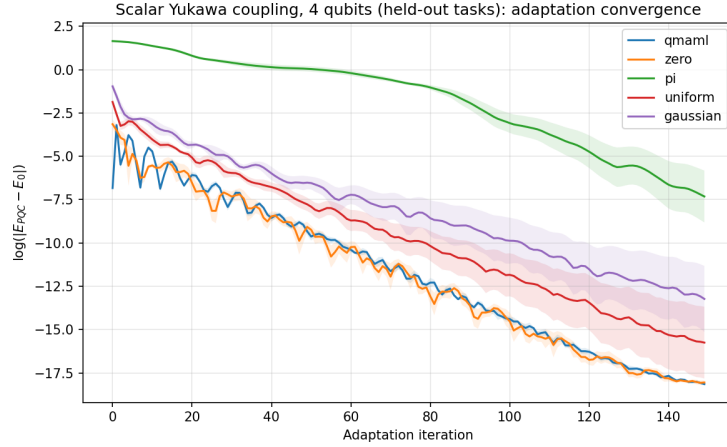


Figure 14: Scalar Yukawa coupling: adaptation convergence by init scheme.

7.10 Track J: Lipkin-Meshkov-Glick model

Flagged as nuclear-structure-adjacent rather than core HEP [27], included for completeness. Precisely differentiated from that source's "HL-VQE": despite the similar name, HL-VQE iteratively co-optimizes an *effective Hamiltonian* via orbital rotation, not a neural-network-learned initialization; we run plain Q-MAML on their own un-rotated Hamiltonian, $J=3/2$ (2 qubits, 4 states, matching their own smallest showcase scale). Verified via the exact $\text{su}(2)$ algebra on the constructed generators and the analytically-known $V=0$ limit.

Genuine null result, not a close call. All five schemes reach the float64 precision floor (log-gap -16.8 to -17.5), and the ranking among them is within run-to-run noise: the convergence trajectories run together in one noisy bundle for the entire 150-iteration budget, never separating, unlike every other track in this paper. Likely explanation: 4 states on 2 qubits may be too small a Hilbert space for a bad initialization to get meaningfully stuck. The clear next step is a *larger* J (more qubits), not more seeds: this benchmark needs to get harder before an initialization-scheme comparison here is informative.

7.11 Cross-track synthesis

Pilot-scale: 2–8 qubits, single seed except where noted, hardware-efficient ansatz throughout except Track H (UCCSD). See §9.

Table 24: Cross-track findings.

Finding
Ansatz depth and pretrain budget, not just init scheme, become load-bearing at larger system size (Track A): under-expressive depth erased Q-MAML’s $L=3$ lead; re-tuning depth alone fixed it. At $L=4$, pretraining itself failed to converge: a different failure mode, not yet resolved.
Track B’s failure is real; one leading mechanism (penalty dominance) is refuted by the V -sweep, not confirmed. What holds: penalty- <i>enforced</i> gauge invariance is worse for Q-MAML than either eliminating gauge structure analytically (A) or having none (C), regardless of penalty strength. Track D confirms this with an independent, non-Abelian gauge theory: gauge eliminated analytically, no penalty term, and Q-MAML wins decisively, again matching Track A rather than Track B.
The zero-parameter fixed point is not a simple Pauli-sum-vs.-Hermitian-matrix property, and is narrower than first thought. It holds exactly on four Pauli-sum Hamiltonians (Heisenberg, molecule, Schwinger, $\mathbb{Z}_2$ LGT, $SU(2)$ LGT) but not on two Hermitian-matrix ones (scalar field, mostly not on SUSY QM), consistent with a Pauli-sum link. Track E breaks that link: its Hamiltonian is a Pauli sum yet the fixed point does not hold, the only such case. Leading (untested) hypothesis: every fixed-point Hamiltonian has a purely- Z mass/diagonal term, while Track E’s mixes Z and X .
No classical scheme is a reliable default across tracks (π : best/2nd on A/B, worst on C/F). Q-MAML is the only consistently competitive scheme, under interpolation only. Track C’s phase-stratified test shows this does not extend to extrapolation across a phase transition, and in one direction Q-MAML starts out worse than task-independent baselines.
Track F: a physics-derived initial state is not automatically better than a generic random one. The source paper’s own hand-picked basis-state init is the worst-starting, close-to-worst-finishing scheme on SUSY QM, despite being chosen for its ground-state overlap: fidelity at $t=0$ is not the same as a good landscape position for gradient descent given enough adaptation budget.
Track G confirms Track D’s confirmation: a third gauge theory (non-Abelian, un-reduced) with gauge structure eliminated analytically again favors Q-MAML, though more modestly than A or D, and its Zero-init exact fixed point (despite being a dense, non-Pauli-sum Hamiltonian) adds a third distinct Zero-init outcome among Hermitian-matrix Hamiltonians (none/drift/exact on C/F/G), further undermining any clean Pauli-sum-vs.-Hermitian-matrix rule for that pattern.
Not every genuine improvement shows up as a final-value win. Track H (UCCSD, particle-number- conserving by construction) is the first track where every scheme converges to the identical final energy; Q-MAML’s advantage is entirely in convergence <i>shape</i> (a smooth descent vs. a shared oscillatory transient every classical scheme goes through). Track I similarly has Q-MAML tie, not beat, Zero-init, with a clean mechanistic reason (the task range’s ground states are near-vacuum, and Zero-init already starts there). Both are real, positive results for Q-MAML that a same-final-energy-only reading would miss.
Track J is a genuine null result, kept rather than dropped. At $J=3/2$ (2 qubits), no initialization scheme is distinguishable from any other, plausibly because the Hilbert space is too small for a bad initialization to get stuck in. Reported as-is: a benchmark can be too easy to be informative, and that is itself worth knowing before scaling a comparison up.

8 Discussion

Why does the inner-loop-free algorithm beat the GSoC baseline (inner loop + stop-gradient)? Original hypothesis: the stop-gradient trick gives a weaker training signal.

Table 25: Inner-step ablation (2 seeds): meta-loss and validation accuracy vs. inner-loop step count.

Inner steps	0	...	20
Validation accuracy	0.536	...	0.583
Gradient norm	0.13	...	2.49

Tested directly with an ablation sweeping the inner-loop step count from 0 to 20 (Table 25, GSoC baseline otherwise unchanged): final meta-loss is flat across the sweep and validation accuracy *increases* with more inner steps, opposite of the hypothesis. Gradient norm scales up with more inner steps and tracks better, not worse, accuracy. Hypothesis retracted. Also, the GSoC baseline at zero inner steps does not match Q-MAML’s own numbers (meta-loss 0.68 vs. 0.364) even accounting for the known support/query evaluation-split difference between the two implementations: why Q-MAML beats the GSoC baseline remains open.

Does the classification advantage shrink with qubit count, as §6.5 predicts? Partially. A 2-seed pass showed a reversal by 8 qubits; at 4 seeds it does not hold (§6.1, Table 5): the gap trends smaller, directionally consistent, but not confirmed. Solid at every qubit count: the win vs. Gaussian and the GSoC baseline.

“Moderate gradient norm.” The paper frames moderate norm as good conditioning. Q-MAML has the *highest* norm and the best accuracy: the opposite of a monotonic “moderate is best” rule. Does not contradict the underlying point (avoiding vanishing gradients), but “moderate” should not be read as “closer to some middle value is always better.”

[TODO: Attempted: combining Q-MAML with identity-block init [9] to counteract §6.5’s plateau. Construction verified correct (exact identity at init, autodiff + finite differences), but produced exactly-zero gradients everywhere, not the theorem’s predicted non-vanishing ones. Documented separately, omitted from the main results.]

9 Limitations

- **§7 is single-seed** except where noted (Track B’s $V=10$ point, Track C’s interpolation-split headline: both 3-seed). Multi-seed replication of the rest is the top priority for this section.
- **Track B:** the V -sweep refuted the penalty-dominance hypothesis rather than confirming it. Untried: annealing V during pretraining; separating penalty/physical energy into adaptively-weighted loss terms; the structurally gauge-invariant ansatz comparison (the track’s originally intended angle).
- **Track A:** depth re-tune at $L=3$ is two spot-checks (depth 6, 9), not a scan; the $L=4$ failure (pretraining non-convergence) is one spot-check, not resolved. A joint depth $\times$ pretrain-budget $\times$ qubit-count scan is needed.
- **Track C:** phase-stratified test is single-seed; broken $\rightarrow$ symmetric error is large enough (85–97%) that seed noise could reorder the ranking, though “nobody generalizes well here” is likely to hold.
- **Track D:** does not reproduce the source paper’s qubit-reduction trick, its hadron-mass- ratio calculation, or $N=4$ (baryon scale); $N=2$ only, single seed. Its unusually clean result is itself

a reason to check whether it holds at $N=4$ rather than assume it does.

- **Track E:** the entire premise is a reformulation (ground state instead of the cited literature’s real-time dynamics), see §7.5. Single seed; none of the five schemes had converged within the iteration budget, so the final-value gap may still narrow with more iterations, as it did in §6.3.
- **Track F:** single seed; 9 held-out tasks split unevenly across three superpotentials by random draw (2–4 each), not a balanced/stratified split; uses a fixed-depth hardware-efficient ansatz, not the source paper’s adaptive-VQE operator-selection approach.
- **Track G:** does not reproduce the source paper’s qubit-reduced (matter-removal) Hamiltonian or its trapped-ion qudit gate set, since several of that reduction’s intermediate equations could not be extracted with full confidence; uses the original, un-reduced Hamiltonian at $N=2$ instead. Single seed; no plaquette term exists in a 1D chain at any N .
- **Track H:** $N=2$ for all three nuclei, single seed, task parameters (t, C, D) swept continuously rather than fixed to the source paper’s own experimentally-calibrated values. The `warm_start_nearby` baseline approximates, not reproduces, the source paper’s own warm-start protocol.
- **Track I:** single seed; task range deliberately narrowed to the boson truncation’s accurate regime, meaning this specifically tests the near-vacuum corner of the model’s parameter space, plausibly why Zero-init is competitive here unlike elsewhere in this paper.
- **Track J:** single seed, $J=3/2$ only; no comparison against HL-VQE’s own reported numbers was attempted, and given the null result at this scale, a larger J is needed before such a comparison would be meaningful either way.
- **Ansatz:** nine of the ten tracks use the same hardware-efficient ansatz, not the symmetry-preserving circuits available for several of them (A/B/D/G); Track H uses UCCSD instead, the paper’s only ansatz-family comparison. Isolates init-scheme comparison from ansatz-design confounds on nine tracks, but results there are about relative init quality on a fixed ansatz, not best achievable VQE performance.
- **CPU-only simulation** caps qubit count/depth (14 for the cheap VQE barren-plateau scan, 8 for classification) below where plateaus are most severe on real hardware or GPU-scale simulation.
- **4 seeds** (HIGGS classification) resolves significance vs. Gaussian/GSoC baseline and catches the 2-seed false positive, but not the vs.-best-classical comparison ($p = 0.08\text{--}0.29$). 8–10 seeds is the next step. **Quark-gluon classification is 2 seeds only** (§6.1): the ~ 18 -point accuracy gap is directionally consistent with HIGGS but not independently significance-tested.
- **Reconstructed Heisenberg ansatz** (paper’s appendix circuit unavailable): the molecule study (paper’s exact ansatz) is the stronger evidence.
- **Task-generation:** single-variable quantile binning untested for sensitivity to bin count or binning variable.
- **Barren-plateau diagnostic scope:** single reference-parameter gradient per sample (standard McClean protocol), not a full Fisher-information/effective-dimension analysis.

10 Conclusion

We tested whether Q-MAML [1] improves PQC trainability in a setting it was never evaluated on (HEP classification), after finding neither prior GSoC attempt actually ran the published algorithm. Our system matches the paper’s own VQE results (reconstructed and exact-ansatz settings) before extending to classification, where it beats classical init and the prior codebase with statistical significance at every qubit count tested. A barren-plateau scan bounds the claim: the benefit is

landscape placement, not plateau resistance, and classification plateaus faster than the VQE setting the paper was validated on.

Beyond classification, a pilot on ten lattice/field-theory (and adjacent) Hamiltonians (§7), what the GSoC 2026 proposal asks for, gives a mostly-positive, diagnosed result: clear wins on six (Schwinger, ϕ^4 scalar field, SU(2) LGT, collective neutrino oscillations, SUSY QM, $\mathbb{D}_8$ LGT), ties for best on two (light nuclei: same final energy for every scheme, Q-MAML faster; Yukawa: Q-MAML ties Zero-init, both at the precision floor), a genuine null result on one (LMG, too small a system to differentiate any scheme), and worst-scheme on one ($\mathbb{Z}_2$ LGT; Gauss-law penalty implicated but not confirmed as sufficient cause). Track D and Track G (independent Abelian and non-Abelian confirmations of Track A’s gauge-eliminated-analytically pattern), Track F (the source paper’s own hand-picked physics init loses to generic random init), and Track H (the first ansatz-family comparison in this paper, and the first track where convergence *shape* rather than final value carries the whole result) are the sharpest new results; Tracks E and I required reformulating their cited literature’s question from real-time dynamics to a ground-state search, stated explicitly rather than glossed over. Findings that revise earlier sections: the zero-parameter fixed point is not universal and, per Track E, is not simply a Pauli-sum-vs.-Hermitian-matrix property either; no classical init is a safe default across Hamiltonian families the way Q-MAML nearly is, under interpolation, not extrapolation. Which structural property of a Hamiltonian predicts whether learned initialization helps is now a concrete question with six wins, two ties, one null result, one diagnosed loss, and several partly-tested mechanisms, rather than an open guess. **[TODO: more seeds (8-10) for both classification datasets, qubit counts beyond 8, Track D at $N=4$, Track J at larger J , before submission.]**

Target Venue

(*internal note, remove before submission*) arXiv preprint is the realistic November deliverable. QTML 2026 deadline (30 June 2026) has passed; NeurIPS 2026 workshop timing doesn’t fit November. Plan: post to arXiv, then target QTML 2027 / ML4PS@NeurIPS for peer review.

References

- [1] J. Lee, J. Cho, S. Kim. Q-MAML: Quantum Model-Agnostic Meta-Learning for Variational Quantum Algorithms. *AAAI* 39(17):18137–18144, 2025. arXiv:2501.05906.
- [2] C. Finn, P. Abbeel, S. Levine. Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. *ICML*, 2017.
- [3] A. Cervera-Liarta, J. S. Kottmann, A. Aspuru-Guzik. Meta-Variational Quantum Eigensolver: Learning Energy Profiles of Parameterized Hamiltonians for Quantum Simulation. *PRX Quantum* 2, 020329, 2021.
- [4] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications* 9, 4812, 2018.
- [5] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, P. J. Coles. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications* 12, 1791, 2021.
- [6] Z. Holmes, K. Sharma, M. Cerezo, P. J. Coles. Connecting Ansatz Expressibility to Gradient Magnitudes and Barren Plateaus. *PRX Quantum* 3, 010313, 2022.

- [7] S. Wang et al. Noise-induced barren plateaus in variational quantum algorithms. *Nature Communications* 12, 6961, 2021.
- [8] M. Ragone et al. A Lie algebraic theory of barren plateaus for deep parameterized quantum circuits. *Nature Communications* 15, 7172, 2023.
- [9] E. Grant, L. Wossnig, M. Ostaszewski, M. Benedetti. An initialization strategy for addressing barren plateaus in parametrized quantum circuits. *Quantum* 3, 214, 2019.
- [10] K. Zhang, L. Liu, M.-H. Hsieh, D. Tao. Escaping from the Barren Plateau via Gaussian Initializations in Deep Variational Quantum Circuits. *NeurIPS*, 2022.
- [11] X. Wang et al. Trainability enhancement of parameterized quantum circuits via reduced-domain parameter initialization. arXiv:2302.06858, 2023.
- [12] C. Zhang, L. Jiang, F. Chen. Qracle: A Graph-Neural-Network-based Parameter Initializer for Variational Quantum Eigensolvers. arXiv:2505.01236, 2025.
- [13] C. Zhang, M. Zheng, Q. Lou, F. Chen. DiffQ: Parameter Initialization for Variational Quantum Algorithms via Generative Diffusion Models. arXiv:2509.17324, 2025.
- [14] M. Ait Haddou, M. Belfkir, S. E. El Harrauss. From Qubits to Couplings: A Hybrid Quantum Machine Learning Framework for LHC Physics. arXiv:2511.15672, 2025.
- [15] P. Baldi, P. Sadowski, D. Whiteson. Searching for exotic particles in high-energy physics with deep learning. *Nature Communications* 5, 4308, 2014.
- [16] Trapped-ion demonstration of variational quantum simulation of the interacting Schwinger model. arXiv:2504.20824, 2025.
- [17] Barren-plateau-free variational quantum simulation of $\mathbb{Z}_2$ lattice gauge theories. arXiv:2507.19203, 2025.
- [18] A. C. Y. Li, A. Macridin, S. Mrenna, P. Spentzouris. Simulating scalar field theories on quantum computers with limited resources. FERMILAB-PUB-22-757-QIS, arXiv:2210.07985, 2023.
- [19] Quantum data learning for quantum simulations in high-energy physics. arXiv:2306.17214, 2023.
- [20] SU(2) hadrons on a quantum computer via a variational approach. *Nature Communications* 12, 6499, 2021. arXiv:2102.08920.
- [21] Simulating collective neutrino oscillations on a quantum computer. arXiv:2102.12556, 2021.
- [22] H. Duan, G. M. Fuller, Y.-Z. Qian. Collective Neutrino Oscillations. *Annual Review of Nuclear and Particle Science* 60, 569–594, 2010.
- [23] Simulating supersymmetric quantum mechanics with variational quantum algorithms. arXiv:2603.18749, 2026.
- [24] E. Gaz, P. P. Popov, G. Pardo, M. Lewenstein, P. Hauke, E. Zohar. Quantum Simulation of non-Abelian Lattice Gauge Theories: a variational approach to $\mathbb{D}_8$. arXiv:2501.17863, 2025.
- [25] P. Cifci, S. Akkoyun, L. La Ronde. Systematic VQE Benchmarking of the Deuteron, Triton, and Helium-3 within Lattice Pionless Effective Field Theory. arXiv:2604.20908, 2026.

- [26] T. N. Kaldenbach, M. Heller, G. Alber, V. M. Stojanovic. Digital Quantum Simulation of Scalar Yukawa Coupling. arXiv:2211.02684, 2022.
- [27] C. E. P. Robin, M. J. Savage. Quantum Simulations in Effective Model Spaces (I): Hamiltonian-Learning VQE using Digital Quantum Computers and Application to the Lipkin-Meshkov-Glick Model. *Phys. Rev. C* 108, 024313, 2023. arXiv:2301.05976.